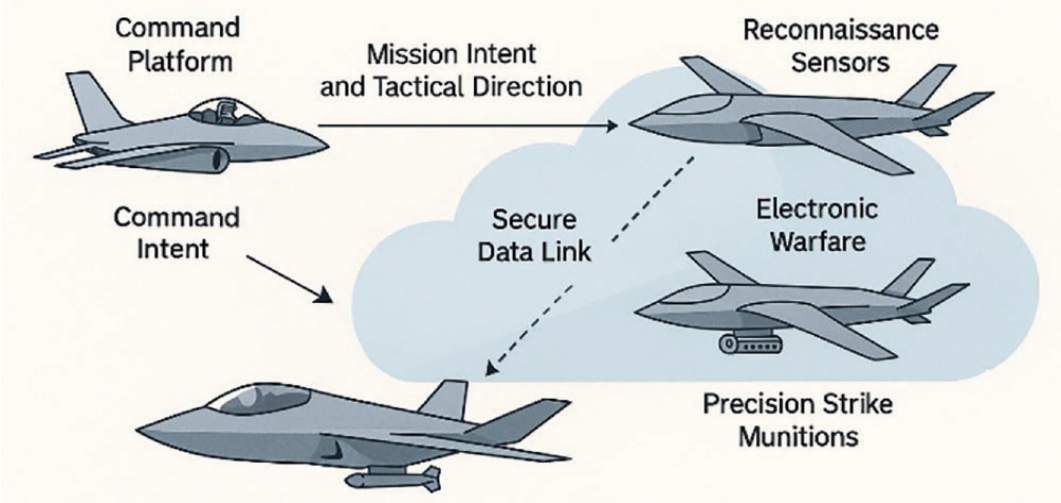


Figure 9.2 Loyal Wingman UAV – human–machine teaming in air combat. ⏎

9.8.3 DJI ActiveTrack – civilian autonomy and real-time object tracking

DJI's ActiveTrack technology represents one of the most widely adopted realtime autonomous tracking systems in the civilian drone sector, demonstrating how AI-driven perception and flight control can make complex aerial operations accessible to non-expert users. ActiveTrack enables a drone to automatically identify, lock onto, and follow a moving subject such as a person, vehicle, animal, or sporting activity while maintaining stable framing and obstacle avoidance. Using computer vision, motion prediction algorithms, and multi-directional sensing, the drone continuously analyses the subject's movement and environment, adjusting its flight path autonomously to keep the target in view (Hansen, et al. 2024).

ActiveTrack operates in multiple tracking modes, including Trace (following behind or alongside a subject), Parallel (flying beside the subject at constant distance), and Spotlight (keeping the camera locked on the subject while the drone moves independently). The system uses deep-learning-based image classification to distinguish the tracked target from background objects and employs onboard sensing such as stereo cameras,